SHI FENG

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EDUCATION

Harvard University *Sept 2023 - Present*
Ph.D. in Computer Science, advised by Prof. Yiling Chen; expected February 2027
Harvard University *Sept 2023 - Feb 2026*
S.M. in Computer Science, GPA: 3.93/4.0
Tsinghua University *Aug 2019 - Jun 2023*
B.E. (Honors) in Computer Science, selected to **Yao Class**, GPA: 3.89/4.0

RESEARCH FOCUS

Large Language Models: post-training, reinforcement learning, reasoning and planning, and evaluation.
Economics and Computation: algorithmic game theory, market design, auctions, and strategic learning.
I study learning from limited supervision, diversity of correct solutions, and mechanisms of model error detection.

APPOINTMENTS

Graduate Research Assistant, Harvard University, advised by Prof. Yiling Chen *Sept 2023 - Present*
Research Intern, Microsoft Research Asia Theory Center, mentored by Prof. Wei Chen *Dec 2022 - Jun 2023*
Research Intern, EconCS Group, Harvard University, mentored by Prof. Yiling Chen *Feb 2022 - Aug 2022*
Research Intern, Microsoft Research Asia Theory Center, mentored by Prof. Wei Chen *Nov 2020 - Jan 2022*

WORKING PAPERS

(* denotes equal contribution or alphabetical order)
Shi Feng, Qimin Zhong, Siwei Wang, Yifei Shen, Shang-Hua Teng, Hao Liao, Wei Chen.
Learning When to Abort in Transformer Planning: A Mechanistic Study.
Under review at **ICLR 2027.** [arXiv]

Hanlin Zhang, **Shi Feng**, Anton Tsitsulin, Vahab Mirrokni, Yiling Chen, Sham M. Kakade.
MOSAIC: Dynamic Benchmarking as Mechanism Design.
Under review at **ICLR 2027.** [PDF]

Yiling Chen*, **Shi Feng***, Sadie Zhao*.
Contextual Procurement Auctions with Bandit Learning.
Under review at **AAAI 2027.** [arXiv]

Shi Feng, Sadie Zhao, Tuo Zhao, Yiling Chen.
When and What to Price against Model Extraction.
Under review at **ICLR 2027.**

PUBLICATIONS

Yiling Chen*, **Shi Feng***, Grant Schoenebeck*, Fang-Yi Yu*.
Information Elicitation by Ordinal Agreement.
International Conference on Web and Internet Economics (WINE 2026). [arXiv]

Shi Feng*, Hanlin Zhang*, Fan Nie, Sham Kakade, Yiling Chen.
Peer-Predictive Self-Training for Language Model Reasoning.
Conference on Language Modeling (COLM 2026). [arXiv]

Siwei Wang*, Yifei Shen*, Haoran Sun*, **Shi Feng***, Shang-Hua Teng, Li Dong, Yaru Hao, Wei Chen.
Benefits and Pitfalls of Reinforcement Learning for Language Model Planning: A Theoretical Perspective.
The Fourteenth International Conference on Learning Representations (ICLR 2026). [arXiv]

Yiling Chen*, **Shi Feng***, Paul Kattuman*, Fang-Yi Yu*.
Data Reliability Scoring.
The Twenty-Seventh ACM Conference on Economics and Computation (EC 2026). [arXiv]

Yiling Chen*, **Shi Feng***, Fang-Yi Yu*.
Carrot and Stick: Eliciting Comparison Data and Beyond.
The 38th Conference on Neural Information Processing Systems (NeurIPS 2024). [arXiv]

Siwei Wang*, Yifei Shen*, **Shi Feng**, Haoran Sun, Shang-Hua Teng, Wei Chen.
ALPINE: Unveiling The Planning Capability of Autoregressive Learning in Language Models.
The 38th Conference on Neural Information Processing Systems (NeurIPS 2024). [arXiv]

Shi Feng*, Nuoya Xiong*, Wei Chen.
Combinatorial Causal Bandits without Graph Skeleton.
The 16th Asian Conference on Machine Learning (ACML 2024). [arXiv]

Shi Feng, Wei Chen.
Combinatorial Causal Bandits.
The 37th AAAI Conference on Artificial Intelligence (AAAI 2023). [arXiv]

Shi Feng, Fang-Yi Yu, Yiling Chen.
Peer Prediction for Learning Agents.
The 36th Conference on Neural Information Processing Systems (NeurIPS 2022). [arXiv]
Shi Feng, Wei Chen.
Causal Inference for Influence Propagation – Identifiability of the Independent Cascade Model.
The 10th International Conference on Computational Data and Social Networks (CSoNet 2021), **Best Paper**. [arXiv]

SELECTED HONORS & AWARDS

Outstanding Undergraduate *Jun 2023*, THU IIIS
"Star of Tomorrow" Award of Excellence *Mar 2022*, MSR Asia
Best Paper Award *(1/57)* *Nov 2021*, CSoNet
"Challenge Cup" Science and Technology Competition, Top Grade Prize *(top 3%)* *Apr 2021*, THU
Tsinghua Xuetang Scholarship *Aug 2019*, THU
Russian Mathematical Olympiad, Silver Medal *Apr 2019*, MSE of Russia
Chinese Mathematical Olympiad, Gold Medal *Nov 2018*, CMS

TALKS

ESIF-AIML 2026, Data Reliability Scoring *Jun 2026*
Harvard EconCS Seminar, Data Reliability Scoring *Nov 2025*
ACML 2024, Combinatorial Causal Bandits without Graph Skeleton *Dec 2024*
EC-24 Workshop, Carrot and Stick: Eliciting Comparison Data and Beyond *Jul 2024*
Harvard EconCS Seminar, Reveal Planning Capability of Autoregressive Learning in LLMs *Mar 2024*
MSRA Theory Seminar, LLMs: Current Theoretical Analysis and Future Directions *Mar 2023*
MSRA Online Paper Sharing, Combinatorial Causal Bandits *Mar 2023*
AAAI 2023, Combinatorial Causal Bandits *Feb 2023*
NeurIPS 2022, Peer Prediction for Learning Agents *Oct 2022*
Yao Class Seminar, Peer Prediction for Learning Agents *Sept 2022*
CSoNet 2021, Causal Inference for Influence Propagation–Identifiability of the IC Model *Nov 2021*

SERVICES

Journal Reviewer: Journal of Combinatorial Optimization (JOCO), IEEE Journal on Selected Areas in Information Theory (JSAIT), Transactions on Machine Learning Research (TMLR)
Conference Reviewer: ICML 2023–2026, NeurIPS 2023–2026, ICLR 2024–2026,
AAAI 2026/2027, ACML 2025/2027, WINE 2026
Teaching Fellow: Network Science: Theory and Algorithms (2023 Spring) in Tsinghua University, Economics and Computation (2025 Spring), Convex Optimization and Applications in Machine Learning (2026 Spring) in Harvard University